

Evaluation Report — Normal vs Faulty Episodes

H5: /Users/Phips1900/PhD/Research/RelAIBotiX/datasets/IL/act/faults_1k/eval_act_mj2_faults_no1.h5

Episodes: 1000

Success tolerance (XY max-abs): 0.02 m; Target: (0.2, -0.15)

Faulty flags: noise_strength, blur_strength, brightness_strength (episode max > 0)

Summary — success rates & episode stats

subset	n_episodes	success_rate	mean_pos_err	median_pos_err	avg_duration sec	avg_n_samples
blur_only	291	60.14	0.073866	0.014080	9.992509	1000.250859
brightness_only	298	72.82	0.057965	0.009245	9.992550	1000.255034
combined	0	NaN	NaN	NaN	NaN	NaN
faulty_any	900	56.11	0.080876	0.014145	9.992667	1000.266667
noise_only	311	36.33	0.109387	0.084971	9.992926	1000.292605
normal	100	88.00	0.029393	0.007273	9.996400	1000.640000

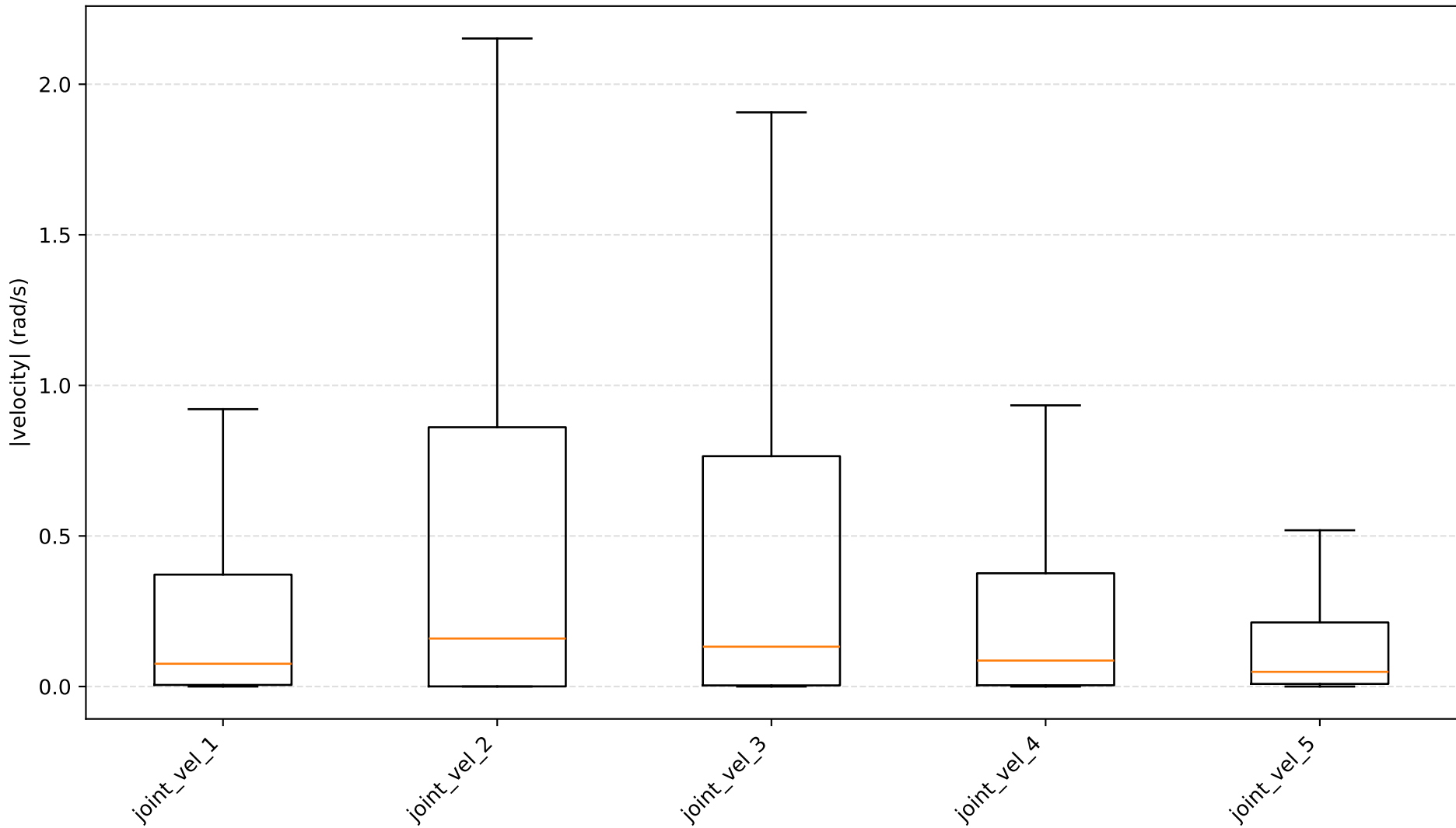
Per-episode (HEAD)

run_id	start_idx	end_idx	n_samples	duration_sec	success	pos_err	goal_x	goal_y	final_x	final_y	noise_strength_max	blur_strength_max
0	0	999	1000	9.99	True	0.013449	0.2	-0.15	0.200010	-0.136551	0.0	0.0
1	1000	2000	1001	10.00	True	0.005435	0.2	-0.15	0.194565	-0.153346	0.0	0.0
2	2001	3000	1000	9.99	True	0.004882	0.2	-0.15	0.195709	-0.154882	0.0	0.0
3	3001	4000	1000	9.99	True	0.004568	0.2	-0.15	0.203659	-0.154568	0.0	0.0
4	4001	5000	1000	9.99	True	0.003525	0.2	-0.15	0.203473	-0.153525	0.0	0.0
5	5001	6001	1001	10.00	True	0.007453	0.2	-0.15	0.207453	-0.151864	0.0	0.0
6	6002	7002	1001	10.00	True	0.007486	0.2	-0.15	0.207486	-0.149019	0.0	0.0
7	7003	8003	1001	10.00	True	0.004195	0.2	-0.15	0.203249	-0.154195	0.0	0.0
8	8004	9004	1001	10.00	True	0.001600	0.2	-0.15	0.198400	-0.149102	0.0	0.0
9	9005	10005	1001	10.00	True	0.010404	0.2	-0.15	0.210404	-0.149990	0.0	0.0
10	10006	11005	1000	9.99	True	0.010793	0.2	-0.15	0.210793	-0.143973	0.0	0.0
11	11006	12005	1000	9.99	True	0.001575	0.2	-0.15	0.199413	-0.151575	0.0	0.0
12	12006	13005	1000	9.99	True	0.003028	0.2	-0.15	0.196972	-0.149511	0.0	0.0
13	13006	14005	1000	9.99	True	0.013516	0.2	-0.15	0.213516	-0.160908	0.0	0.0
14	14006	15005	1000	9.99	True	0.018554	0.2	-0.15	0.196942	-0.168554	0.0	0.0
15	15006	16005	1000	9.99	True	0.002131	0.2	-0.15	0.197869	-0.150737	0.0	0.0
16	16006	17005	1000	9.99	True	0.002823	0.2	-0.15	0.202823	-0.147378	0.0	0.0
17	17006	18005	1000	9.99	True	0.011104	0.2	-0.15	0.211104	-0.151781	0.0	0.0
18	18006	19005	1000	9.99	True	0.005160	0.2	-0.15	0.197746	-0.155160	0.0	0.0
19	19006	20005	1000	9.99	False	0.038685	0.2	-0.15	0.222257	-0.188685	0.0	0.0
20	20006	21005	1000	9.99	True	0.002928	0.2	-0.15	0.197072	-0.151800	0.0	0.0
21	21006	22005	1000	9.99	True	0.008556	0.2	-0.15	0.208556	-0.151988	0.0	0.0
22	22006	23005	1000	9.99	True	0.005256	0.2	-0.15	0.194744	-0.147489	0.0	0.0
23	23006	24005	1000	9.99	True	0.004590	0.2	-0.15	0.199278	-0.154590	0.0	0.0
24	24006	25005	1000	9.99	True	0.013963	0.2	-0.15	0.213963	-0.158016	0.0	0.0
25	25006	26005	1000	9.99	True	0.002196	0.2	-0.15	0.199160	-0.147804	0.0	0.0
26	26006	27005	1000	9.99	False	0.321240	0.2	-0.15	0.174584	0.171240	0.0	0.0
27	27006	28005	1000	9.99	True	0.007073	0.2	-0.15	0.192927	-0.152852	0.0	0.0
28	28006	29005	1000	9.99	True	0.008074	0.2	-0.15	0.205181	-0.141926	0.0	0.0
29	29006	30005	1000	9.99	True	0.004240	0.2	-0.15	0.195760	-0.153737	0.0	0.0

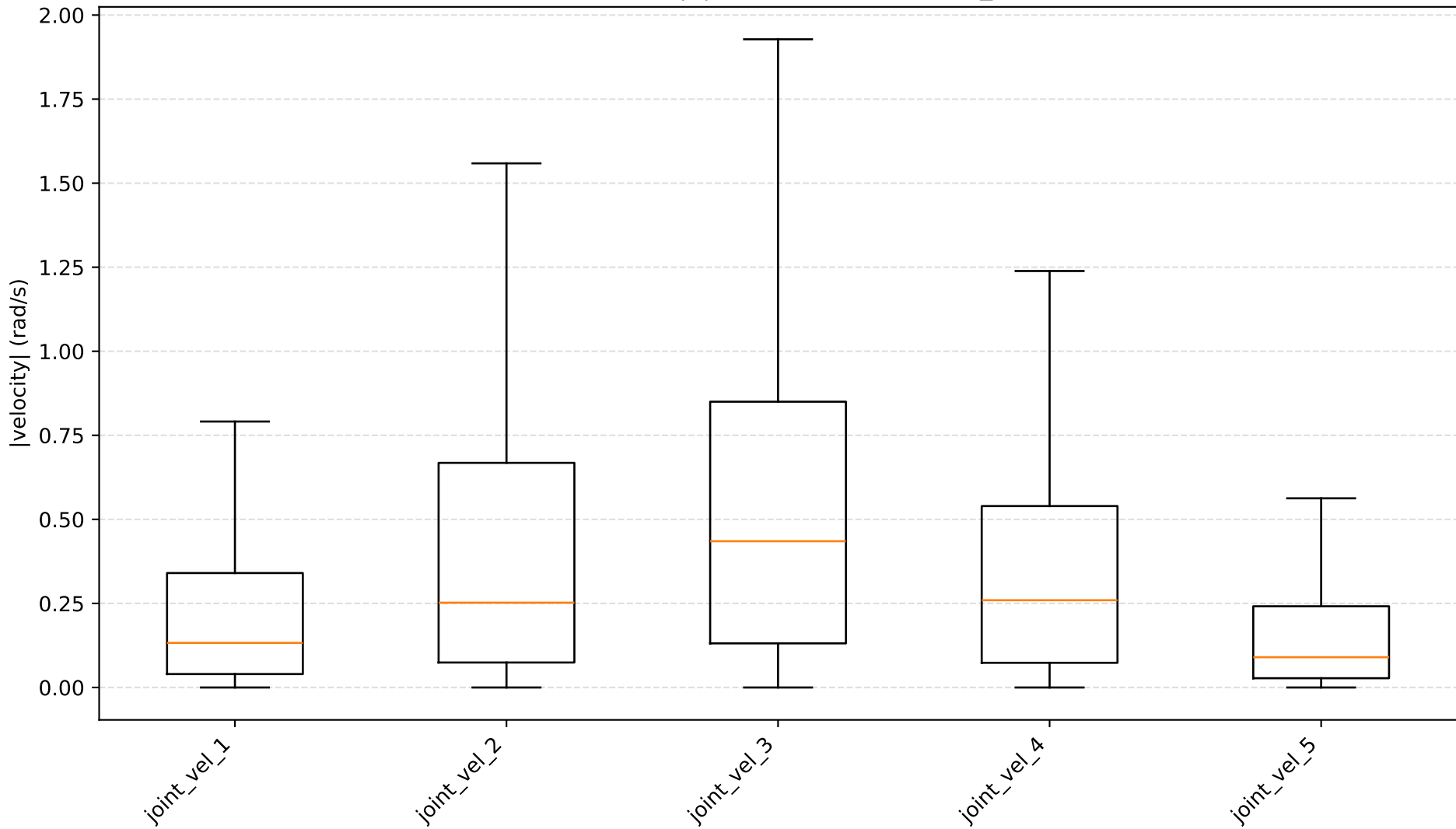
Per-episode (TAIL)

run_id	start_idx	end_idx	n_samples	duration_sec	success	pos_err	goal_x	goal_y	final_x	final_y	noise_strength_max	blur_strength_max
970	970304	971303	1000	9.99	True	0.002760	0.2	-0.15	0.202760	-0.152610	0.030443	0.000000
971	971304	972303	1000	9.99	False	0.151702	0.2	-0.15	0.205875	0.001702	0.000000	0.850171
972	972304	973303	1000	9.99	False	0.180891	0.2	-0.15	0.225265	0.030891	0.000000	0.560131
973	973304	974303	1000	9.99	True	0.001523	0.2	-0.15	0.198477	-0.150799	0.000000	0.273131
974	974304	975303	1000	9.99	True	0.017830	0.2	-0.15	0.214326	-0.167830	0.000000	0.000000
975	975304	976303	1000	9.99	True	0.008769	0.2	-0.15	0.200775	-0.141231	0.000000	0.000000
976	976304	977303	1000	9.99	False	0.236332	0.2	-0.15	0.233000	0.086332	0.200878	0.000000
977	977304	978303	1000	9.99	False	0.235969	0.2	-0.15	0.189754	0.085969	0.000000	0.831900
978	978304	979303	1000	9.99	True	0.004679	0.2	-0.15	0.203377	-0.154679	0.092955	0.000000
979	979304	980303	1000	9.99	False	0.133352	0.2	-0.15	0.262354	-0.016648	0.000000	0.596100
980	980304	981303	1000	9.99	True	0.004578	0.2	-0.15	0.204578	-0.152094	0.000000	0.000000
981	981304	982303	1000	9.99	False	0.185327	0.2	-0.15	0.220221	0.035327	0.965773	0.000000
982	982304	983303	1000	9.99	True	0.004037	0.2	-0.15	0.198157	-0.154037	0.000000	0.000000
983	983304	984303	1000	9.99	False	0.267640	0.2	-0.15	0.223860	0.117640	0.000000	0.382831
984	984304	985303	1000	9.99	False	0.045983	0.2	-0.15	0.191326	-0.104017	0.000000	0.908760
985	985304	986303	1000	9.99	False	0.090766	0.2	-0.15	0.284216	-0.059234	0.000000	0.858151
986	986304	987303	1000	9.99	False	0.293028	0.2	-0.15	0.206527	0.143028	0.000000	0.000000
987	987304	988303	1000	9.99	False	0.066252	0.2	-0.15	0.266252	-0.119782	0.825634	0.000000
988	988304	989303	1000	9.99	True	0.003934	0.2	-0.15	0.203934	-0.150791	0.000000	0.000000
989	989304	990303	1000	9.99	True	0.003113	0.2	-0.15	0.202634	-0.146887	0.378226	0.000000
990	990304	991303	1000	9.99	True	0.007619	0.2	-0.15	0.205027	-0.157619	0.427115	0.000000
991	991304	992303	1000	9.99	True	0.002860	0.2	-0.15	0.201455	-0.152860	0.000000	0.018260
992	992304	993303	1000	9.99	False	0.119580	0.2	-0.15	0.284196	-0.030420	0.483610	0.000000
993	993304	994303	1000	9.99	False	0.207155	0.2	-0.15	0.240848	0.057155	0.000000	0.921711
994	994304	995303	1000	9.99	True	0.005630	0.2	-0.15	0.198879	-0.144370	0.000000	0.011270
995	995304	996303	1000	9.99	False	0.292575	0.2	-0.15	0.244280	0.142575	0.773135	0.000000
996	996304	997303	1000	9.99	True	0.004249	0.2	-0.15	0.199736	-0.145751	0.000000	0.000000
997	997304	998303	1000	9.99	False	0.165063	0.2	-0.15	0.194849	0.015063	0.819607	0.000000
998	998304	999303	1000	9.99	True	0.003374	0.2	-0.15	0.203374	-0.152316	0.000000	0.000000
999	999304	1000303	1000	9.99	True	0.008966	0.2	-0.15	0.208966	-0.157003	0.051721	0.000000

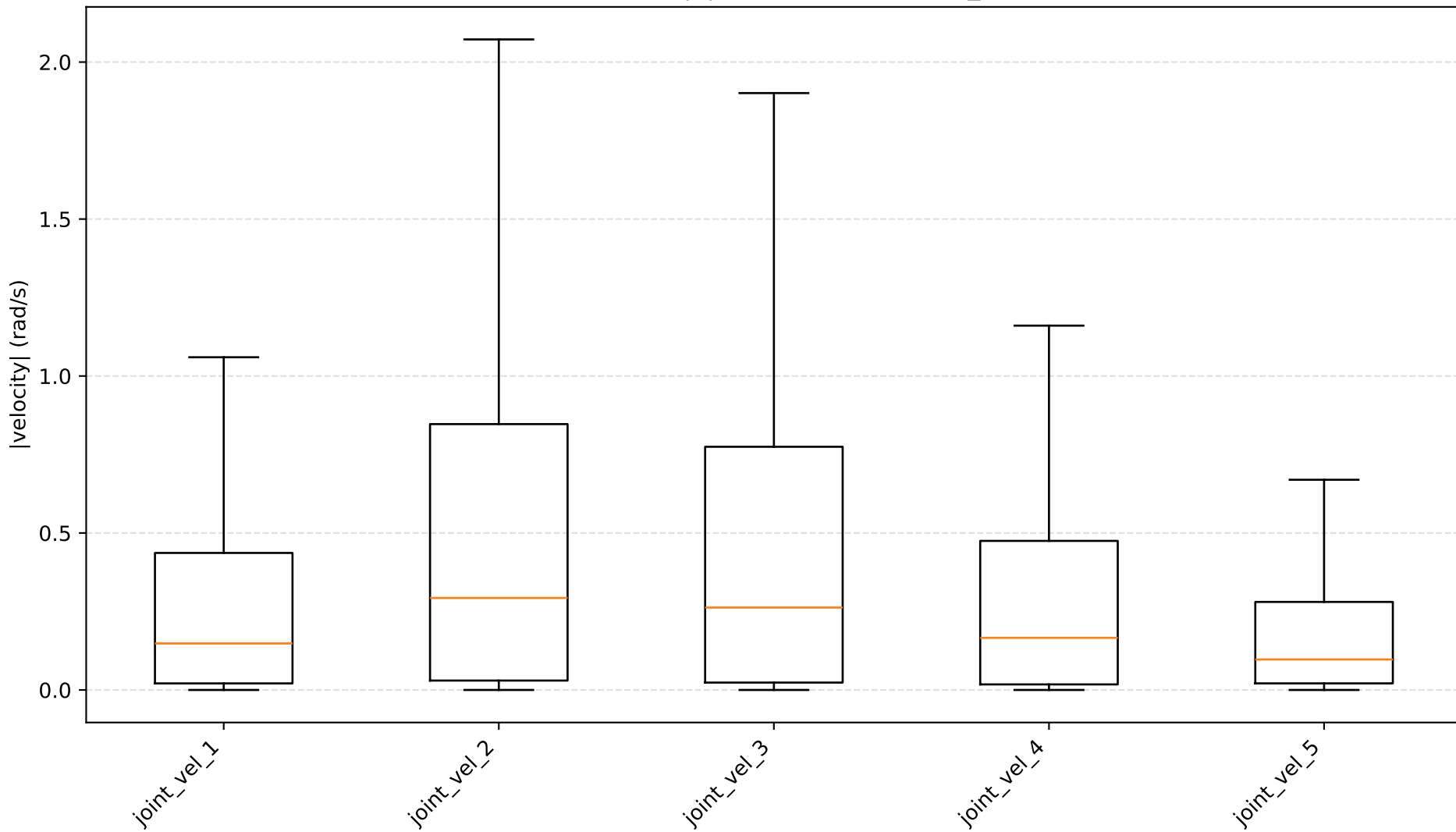
Joint velocity $|v|$ distributions — normal



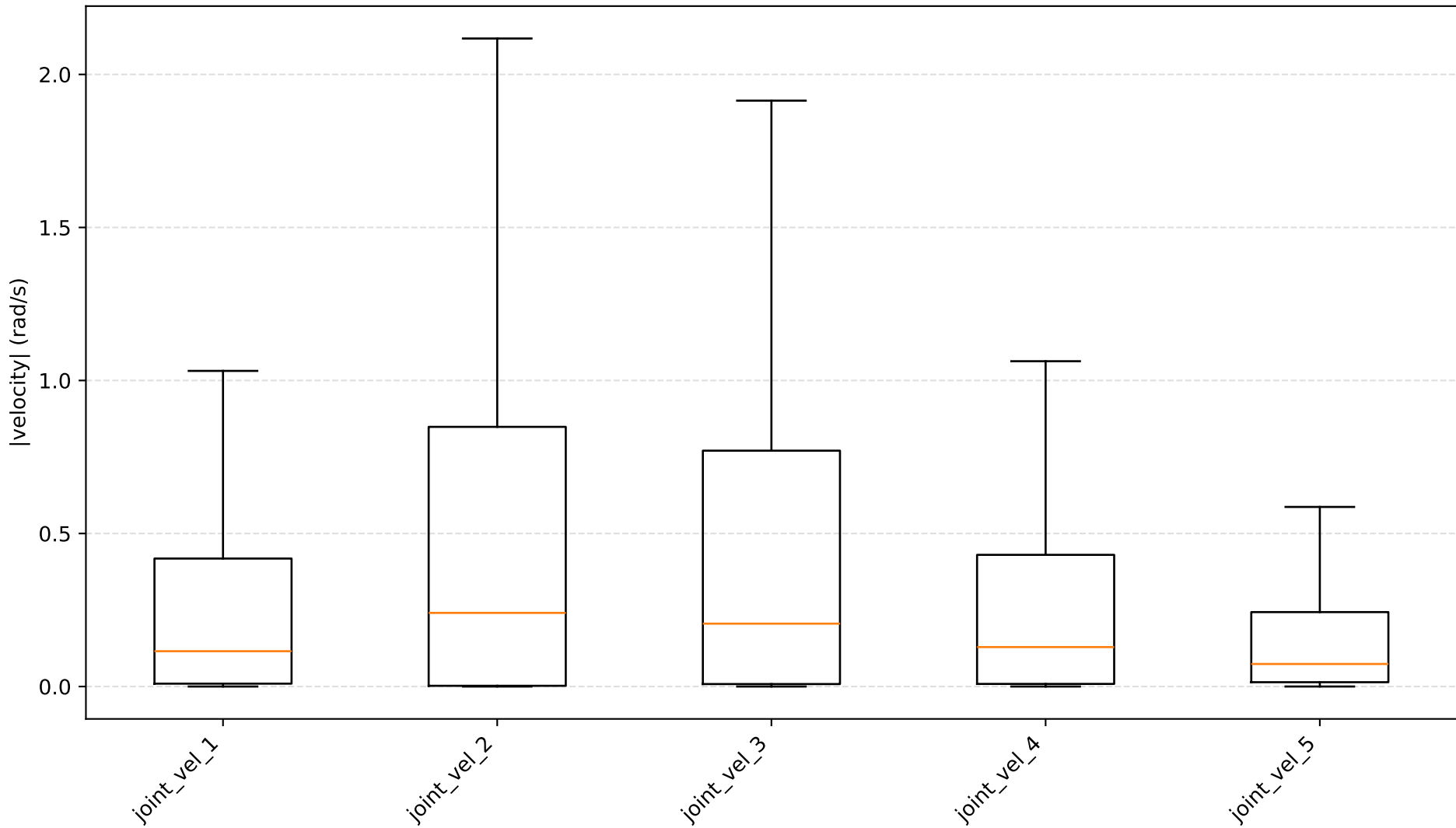
Joint velocity $|v|$ distributions — noise_only



Joint velocity $|v|$ distributions — blur_only



Joint velocity $|v|$ distributions — brightness_only



Joint velocity $|v|$ distributions — combined

